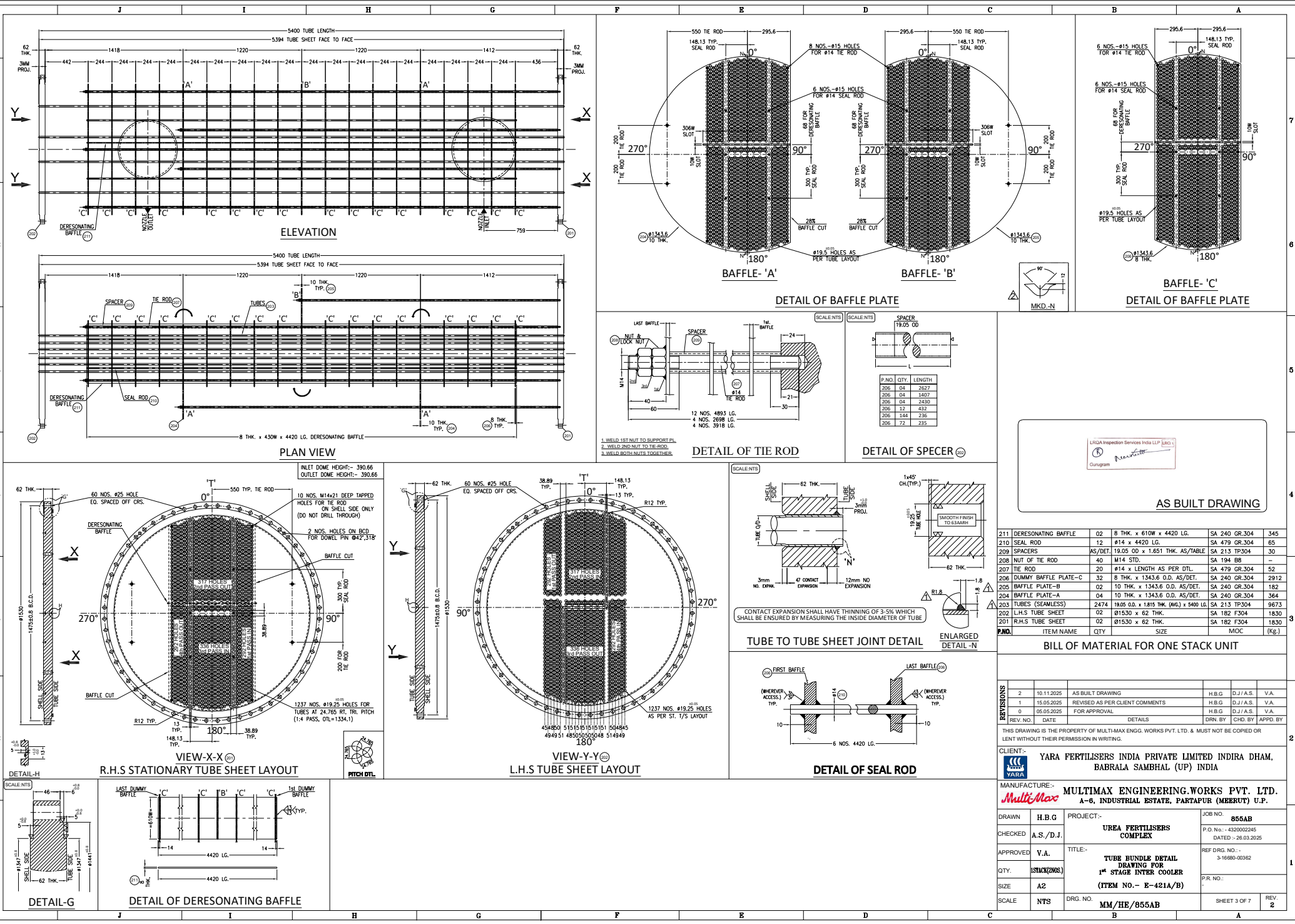
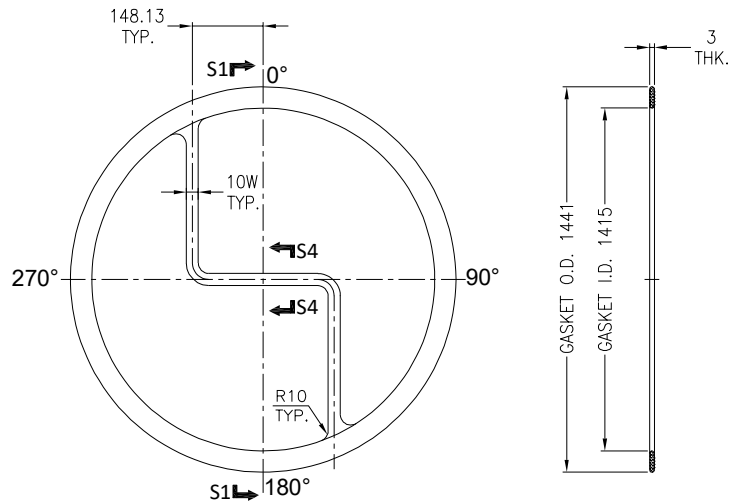
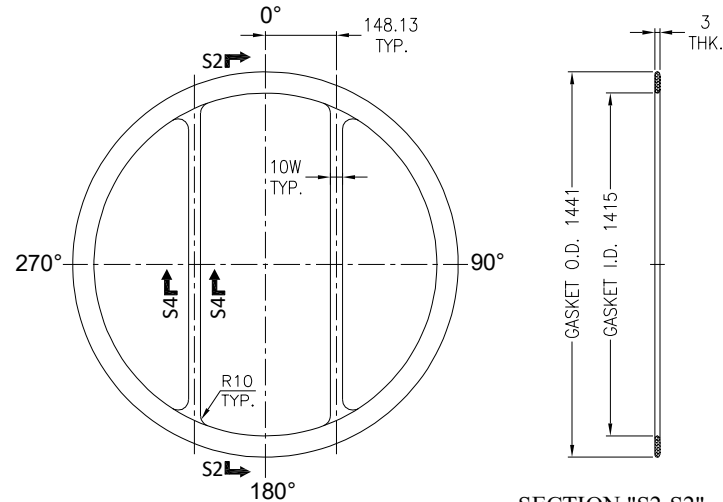






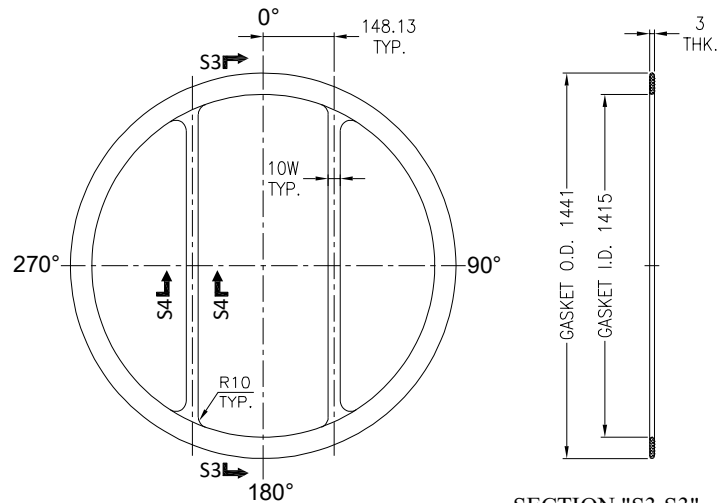
SECTION "S1-S1"

GASKET DETAIL FOR (401)
LHS CH. FLG. TO LHS TUBESHEET



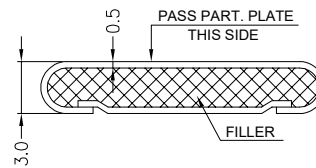
SECTION "S2-S2"

GASKET DETAIL FOR (402)
RHS TUBESHEET TO RHS CH. FLG.



SECTION "S3-S3"

GASKET DETAIL FOR (403)
RHS CH. COVER FLG. TO CH. COVER



SECTION VIEW "S4-S4"



AS BUILT DRAWING

(*) IRAN JACKETED ARAMIDE FILLER

P. NO	DESCRIPTION	QTY	SIZE	MATERIAL	WT (kg)
403	GASKET FOR RHS CH. COVER FLG. TO CH. COVER	2+4+4	3 THK. x 1441 OD x 1415 ID	UA (*)	-
402	GASKET FOR RHS T/S TO RHS CH. FLG.	2+4+4	3 THK. x 1441 OD x 1415 ID	UA (*)	-
401	GASKET FOR LHS CH. FLG. TO LHS T/S	2+4+4	3 THK. x 1441 OD x 1415 ID	UA (*)	-

BILL OF MATERIAL FOR ONE STACK UNIT

REVISIONS	REV. NO.	DATE	DETAILS	DRN. BY	CHD. BY	APPO. BY
1	10.11.2025	AS BUILT DRAWING		H.B.G	D.J./A.S.	V.A.
0	05.05.2025	FOR APPROVAL		M.K	D.J./A.S.	V.A.

THIS DRAWING IS THE PROPERTY OF MULTIMAX ENGG. WORKS PVT. LTD. & MUST NOT BE COPIED OR LENT WITHOUT THEIR PERMISSION IN WRITING.

CLIENT:- YARA FERTILISERS INDIA PRIVATE LIMITED INDIRA DHAM, BABRALA SAMBHAL (UP) INDIA

MANUFACTURE:- MULTIMAX ENGINEERING WORKS PVT. LTD. A-6, INDUSTRIAL ESTATE, PARTAPUR (MEERUT) U.P.

DRAWN	H.B.G	PROJECT:-	JOB NO.
CHECKED	A.S./D.J.	UREA FERTILISERS COMPLEX	855AB
APPROVED	V.A.	TITLE:-	P.O. No.:- 432002245
QTY.	STACK(205)	GASKET DETAIL DRAWING FOR 1 st STAGE INTER COOLER	DATED :- 26.03.2025
SIZE	A2	(ITEM NO.- E-421A/B)	REF DRG. NO.:-
SCALE	NTS	DRG. NO.	3-16680-00362
		MM/HE/855AB	P.R. NO.:-
			-
			SHEET 4 OF 7
			REV. 1

(A) GENERAL

- (B) MATERIAL

- ### (C) FABRICATION

- (D) HYDROTEST

- (E) PROTECTION & DISPATCHED

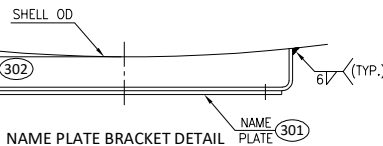
- HEAT TREATMENT CYCLE FOR DISHED END:-

PAINTING:—

01. EXTERNAL PAINTING C.S. MATERIAL:-

MANDATORY SPARES :-

- MANUFACTURER NAME PLATE DETAIL (301)



- 1) ALL DIMENSIONS ARE IN M.M. UNLESS OTHERWISE SPECIFIED.
- 2) ALL LETTERS AND LETTER BLOCKS SHALL BE ENGRAVED IN BLACK.
- 3) NAME PLATE SHALL BE RIVETED WITH 50 ALUMINIUM RIVETS AND SHALL BE TACK WELDED TO BRACKET ON ALL FOUR SIDES.
- 4) ***DATE OF TEST SHALL BE ADDED LATER ON AS PER ACTUAL.
- 5) SEPARATE NAME PLATE SHALL BE PROVIDED FOR BOTH TAG.

AS BUILT DRAWING

BILL OF MATERIAL FOR ONE STACK UNIT

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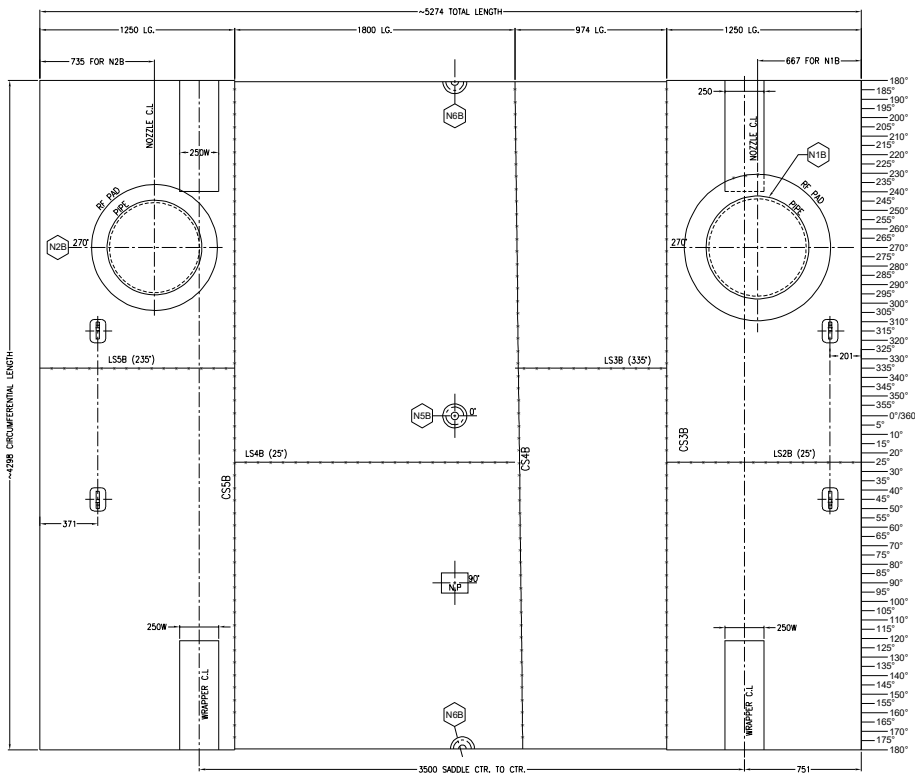
CLIENT:-

CLIENT:-  YARA FERTILISERS INDIA PRIVATE LIMITED INDIRA DHAM,
BABRALA SAMBHAL (UP) INDIA

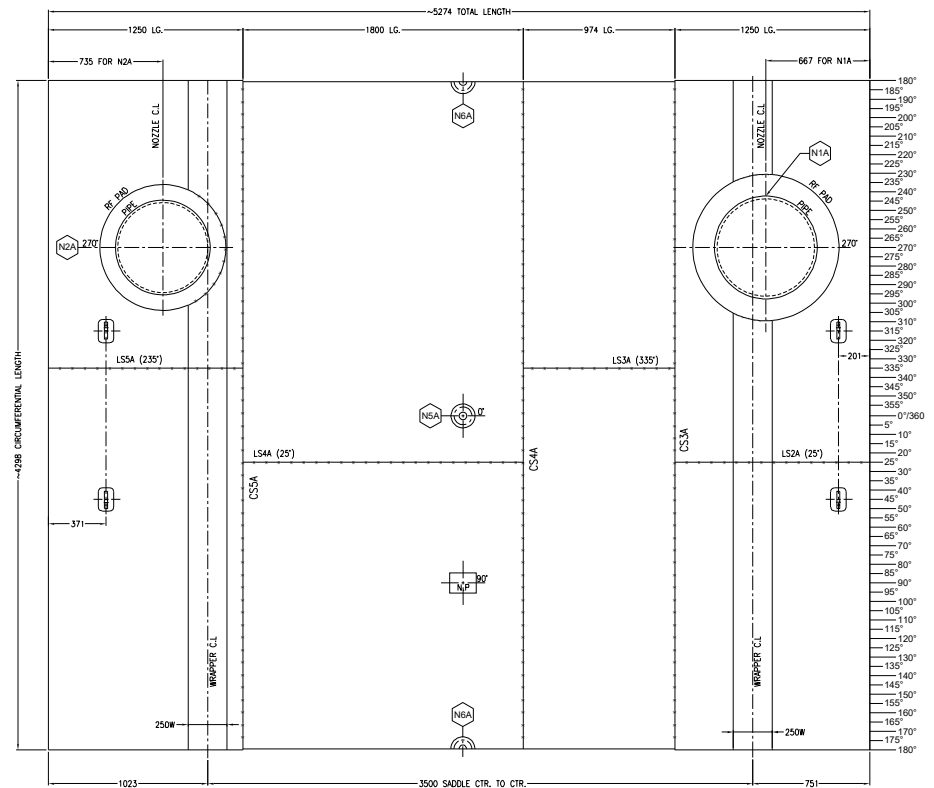
MANUFACTURE:-

MANUFACTURE:- **MULTIMAX ENGINEERING WORKS PVT. LTD.**
A-6, INDUSTRIAL ESTATE, PARTAPUR (MEERUT) U.P.

DRAWN	H.B.G	PROJECT:-	JOB NO. 855AB
CHECKED	A.S./D.J.	UREA FERTILISERS COMPLEX	P.O. NO. - 4320002245 DATED - 26.03.2025
APPROVED	V.A.	TITLE:-	REF DRG. NO. - 3-16680-03062
QTY. 1STUCK(2000)		GENERAL NOTES & N.P DETAIL DRAWING FOR 1ST STAGE INTER COMPLEX (ITEM NO. ~ (R-421A/B)	P.R. NO. -
SIZE A2			
SCALE NTS	DRG. NO.	MM/HE/855AB	SHEET 5 OF 7 REV. 3



SHELL DEVELOPMENT FOR TOP EXCHANGER



SHELL DEVELOPMENT FOR BOTTOM EXCHANGER

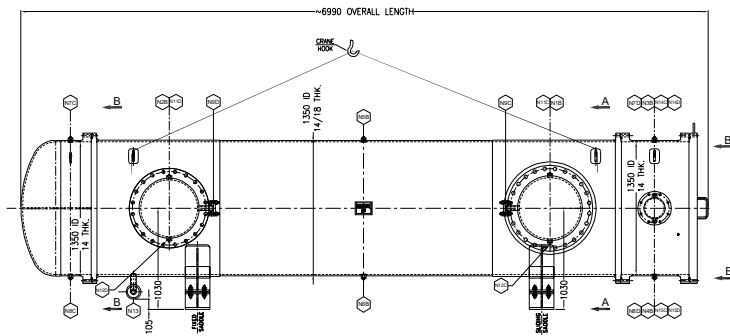
GENERAL NOTES:-

- ALL DIMENSIONS ARE IN MILLIMETERS.
- ALL WELDING SHALL BE GROUND FLUSH. WHEREVER WELDING IS FOULING WITH OTHER WELD.

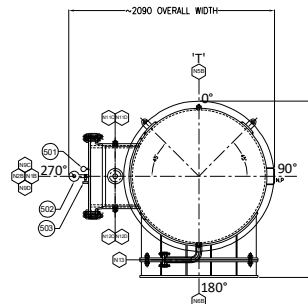


AS BUILT DRAWING

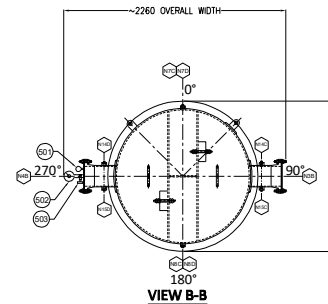
REV. NO.	DATE	DETAILS	DRN. BY	CHD. BY	APPD. BY
2	10.11.2025	AS BUILT DRAWING	H.B.G	D.J./A.S.	V.A.
1	15.05.2025	REVISED AS PER CLIENT COMMENTS	H.B.G	D.J./A.S.	V.A.
0	05.05.2025	FOR APPROVAL	H.B.G	D.J./A.S.	V.A.
THIS DRAWING IS THE PROPERTY OF MULTIMAX ENGG. WORKS PVT. LTD. & MUST NOT BE COPIED OR LENT WITHOUT THEIR PERMISSION IN WRITING.					
CLIENT:- YARA FERTILISERS INDIA PRIVATE LIMITED INDIRA DHAM, BABRALA SAMBHAL (UP) INDIA					
MANUFACTURE:- MULTIMAX ENGINEERING WORKS PVT. LTD. A-6, INDUSTRIAL ESTATE, PARTAPUR (MEERUT) U.P.					
DRAWN	H.B.G	PROJECT:-	JOB NO. 855AB		
CHECKED	A.S./D.J.	UREA FERTILISERS COMPLEX	P.O. No.:- 4320002245 DATED :- 26.03.2025		
APPROVED	V.A.	TITLE:-	REF DRG. NO.:- 3-16880-00362		
QTY.	1STACK(2NO.)	SHELL DEVELOPMENT DETAIL DRAWING FOR 1 ST STAGE INTER COOLER (ITEM NO.- E-421A/B)	P.R. NO.:-		
SIZE	A2	DRG. NO. MM/HE/855AB	SHEET 6 OF 7		
SCALE	NTS		REV. 2		



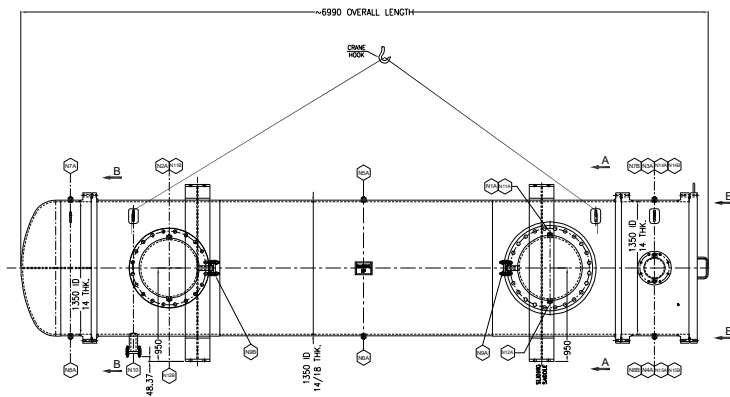
ELEVATION FOR TAG NO.- E-421B



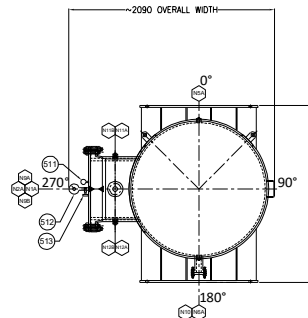
VIEW A-A
SHELL SIDE NOZZLE ORIENTATION



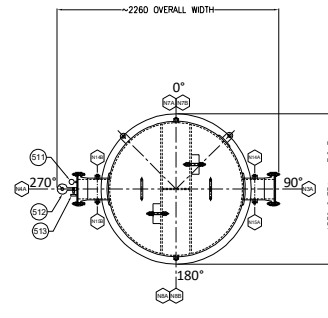
VIEW B-B
LHS & RHS CHANNEL SIDE
NOZZLE ORIENTATION



ELEVATION FOR TAG NO.- E-421A



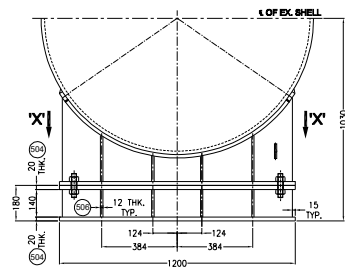
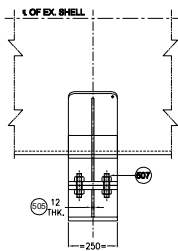
VIEW A-A
SHELL SIDE NOZZLE ORIENTATION



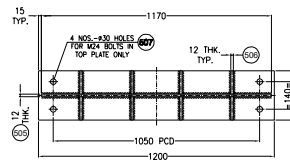
VIEW B-B
LHS & RHS CHANNEL SIDE
NOZZLE ORIENTATION

TRANSPORTATION SADDLE FOR TAG NO.- E-421B

ALL FILLET WELD SIZE SHALL BE 10mm



VIEW H-H
SADDLE DETAIL



VIEW-X-X
TOP & BOTTOM PLATE

				TOTAL: 80 Kg.	
516	STUD + 2 HEX NUT FOR NOZZLE COVER PLATE	16	AS PER REQ. (MIN. 4 NOS / NOZZLE)	M.S.	---
515	NOZZLE COVER PLATE FOR NOZZ. N3A & N4A	02	AS PER NOZZLE (MIN. 10 THK.)	IS 2062	---
514	NOZZLE COVER PLATE FOR NOZZ. N1A & N2A	02	AS PER NOZZLE (MIN. 20 THK.)	IS 2062	---
513	NEEDLE VALVE	02	---	---	---
512	PRESSURE GAUGE	02	---	---	---
511	NON RETURN VALVE	02	---	---	---

BILL OF MATERIAL FOR TAG NO.- E-421A

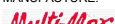
				TOTAL: 300 Kg.	
510	STUD + 2 HEX NUT FOR NOZZLE COVER PLATE	16	AS PER REQ. (MIN. 4 NOS / NOZZLE)	M.S.	---
509	NOZZLE COVER PLATE FOR NOZZ. N3B & N4B	02	AS PER NOZZLE (MIN. 10 THK.)	IS 2062	---
508	NOZZLE COVER PLATE FOR NOZZ. N1B & N2B	02	AS PER NOZZLE (MIN. 20 THK.)	IS 2062	---
507	STUD + 2 HEX NUT FOR TRANSPORTATION SADDLE	08	M24 x 120 LG.	M.S.	---
506	RIB PLATE	16	12 THK. x 140 W x 119 LG.	IS 2062	27
505	WEB PLATE	02	12 THK. x 140 W x 1170 LG.	IS 2062	33
504	TOP & BOTTOM PLATE	04	20 THK. x 250 W x 1200 LG.	IS 2062	188
503	NEEDLE VALVE	02	---	---	---
502	PRESSURE GAUGE	02	---	---	---
501	NON RETURN VALVE	02	---	---	---

BILL OF MATERIAL FOR TAG NO.- E-421B

- NOTES:-
1. ALL DIMENSIONS ARE IN MM UNLESS OTHERWISE SPECIFIED.
 2. PRIOR TO SHIPMENT ALL THE FLANGE FACE, THREAD COUPLING, AND OTHER MACHINED COMPONENTS SHALL BE COATED WITH RUST PREVENTIVE. ALL VENT HOLES IN R.F. PADS SHALL BE PLUGGED WITH HARD GREASE.
 3. AFTER COMPLETION OF ALL TESTING AND INSPECTION, THE INSIDE OF COMPLETE EQUIPMENT SHALL BE THOROUGHLY DRAINED AND DRIED OUT.
 4. AFTER DRAINING AND DRYING, THE EQUIPMENT SHALL BE PURGED AND FILLED WITH DRY NITROGEN AT 0.5 Kg/cm²(G) PRESSURE & FITTED WITH A PRESSURE GAUGE AND 1/2" NON RETURN VALVE.
 5. ALL NOZZLE SHALL BE PROVIDED WITH BLIND FLANGES TEMPORARY RUBBER GASKET AND COMPLETE BOLTING. BY APPLYING SUITABLE GREASE, RUST PREVENTIVE OIL OR COATING.
 6. ALL EXPOSED MACHINED SURFACES, SUCH AS FLANGE FACE AND BOLTING SURFACE SHALL BE PROTECTED BY TRANSPORTATION.
 7. ALL LOOSE ITEM SHALL BE SUITABLE PACKED AND MARKED WITH A IDENTIFICATION.
 8. ALL HEAT EXCHANGER APPURTENANCES SUSCEPTIBLE TO DAMAGE SHALL BE SUITABLY PROTECTED DURING TRANSPORTATION.
 9. FRAGILE OR MACHINED COMPONENT SHALL BE SPECIALLY PROTECTED AGAINST NATURE DURING HANDLING AND TRANSIT.
 10. 0, 90, 180 & 270 SHALL BE MARKED ON BOTH SIDE OF THE EQUIPMENT.
 11. CENTER OF GRAVITY IN THE SHOP FABRICATED CONDITION SHALL BE MARKED CLEARLY ON OPPOSITE SIDES OF THE HEAT EXCHANGER.
 12. SLIDING END OF EXCHANGERS SHALL BE PAINTED AS "CAUTION-SLIDING END" IN BLACK IN THE SHOP.
 13. CAUTION MARKS SHALL BE PAINTED ON THE EQUIPMENT NEAR BOLTED BLIND FLANGES TO WARN THE PRESENT OF INERT GAS INSIDE THE EQUIPMENT.
 14. PROPER LASHING AND PACKING SHALL BE PROVIDED BEFORE SHIPPING TO PREVENT THE DAMAGE OF EQUIPMENT.
 15. RUBBER SHEETS SHALL BE PROVIDED BETWEEN LASHING/SHIPPING SADDLE & EQUIPMENT OUTSIDE SURFACE TO PREVENT DAMAGE OF PAINT DURING HANDLING & TRANSPORTATION.
 16. ALL TEMPORARY SHIPPING PART SHALL BE REMOVED AT SITE (BEFORE OPERATION).
 17. SPARE PARTS SHALL BE PACKED SEPARATELY.
 18. SINCE NITROGEN PURGING SHALL BE USED FOR INSIDE OF THE EXCHANGER, PAINT SUITABLE WARNING "EQUIPMENT IS UNDER NITROGEN PURGING" ON VISIBLE PORTION ON OUTSIDE OF SHELL.
 19. BELOW SHIPPING MARK TO BE ADDED IN TRANSPORTATION DRAWING & SHALL BE MARKED IN EQUIPMENT WITH (BOLD WHITE/BLACK PAINT) SUITABLE VISIBLE LETTER HEIGHT.
- CLIENT: YARA FERTILISERS INDIA PRIVATE LIMITED
DESTINATION: M/S YARA FERTILISERS INDIA Pvt. Ltd, INDIRADHAM, BABRALA, DISTT- SAMBHAL (UP), PIN- 24201 (INDIA)
PO NO- 4320002245 DATED: 26.03.2025
EQUIPMENT NO: E-421A/B
EQUIPMENT OVERALL DIMENSION(FOR E-421A): 7150mm(L) x 2410mm(W) x 1950mm(H)
EQUIPMENT OVERALL DIMENSION(FOR E-421B): 7150mm(L) x 2410mm(W) x 1950mm(H)
EQUIPMENT SHIPPING WEIGHT FOR TAG NO.- E-421A: ~13800Kg.
EQUIPMENT SHIPPING WEIGHT FOR TAG NO.- E-421B: ~13600Kg.



AS BUILT DRAWING

REVISIONS									
	1	10.11.2025	AS BUILT DRAWING				H.B.G	D.J./A.S.	V.A.
	0	29.05.2025	FOR APPROVAL				S.S.	D.J./A.S.	V.A.
	REV. NO.	DATE	DETAILS				DRN. BY	CHD. BY	APPD. BY
THIS DRAWING IS THE PROPERTY OF MULTIMAX ENGG. WORKS PVT. LTD. & MUST NOT BE COPIED OR LENT WITHOUT THEIR PERMISSION IN WRITING.									
CLIENT:-									
 YARA FERTILISERS INDIA PRIVATE LIMITED INDIRA DHAM, BABRALA SAMBHAL (UP) INDIA									
MANUFACTURE:-									
 MULTIMAX ENGINEERING.WORKS PVT. LTD. A-6, INDUSTRIAL ESTATE, PARTAPUR (MEERUT) U.P.									
DRAWN	S.S.	PROJECT:-				JOB NO. 855AB			
CHECKED	A.S./D.J.	UREA FERTILISERS COMPLEX				P.O. No.- 4320002245 DATED :- 26.03.2025			
APPROVED	V.A.	TITLE:-				REF DRG. NO. :-			
QTY.	1STACK(2NOS)	TRANSPORTATION DETAIL DRAWING FOR 1 st STAGE INTER COOLER (ITEM NO.- E-421A/B)				P.R. NO. :-			
SIZE	A2								
SCALE	NTS	DRG. NO. MM/HE/855AB				SHEET 7 OF 7			
								REV. 1	